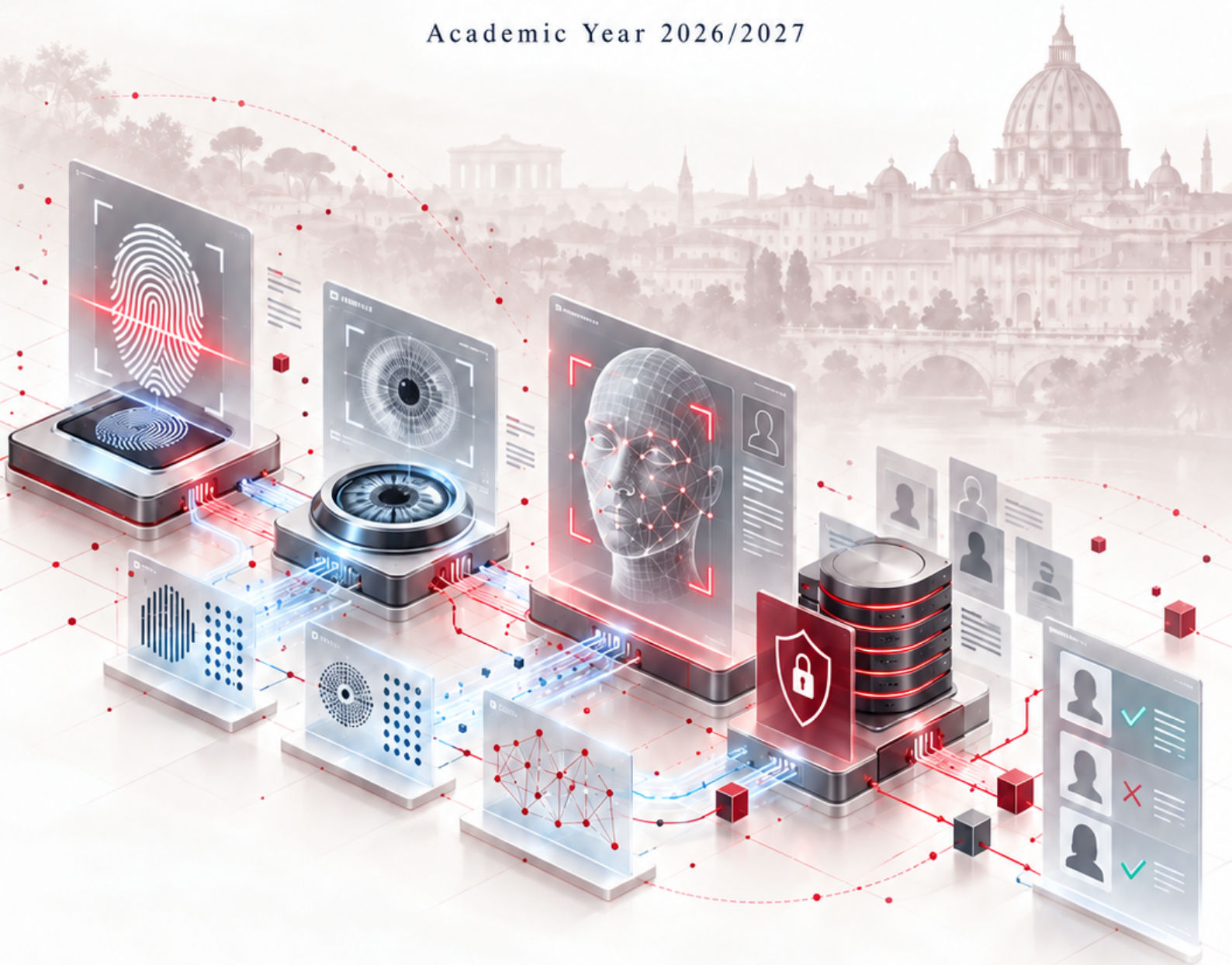
Master's Degree in Computer Science

BIOMETRIC SYSTEMS

Course Notes

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Lesson 0 — Course Overview

The introductory course deck establishes the scope of *Biometric Systems* and the way the course is organized. Administrative information such as the current classroom, schedule, and meeting times is deliberately left in the source-dependent form "see the updated course page", because those details can change from one academic year to another.

Course Scope

The term *biometrics* comes from the Greek words *bios* (life) and *métron* (measure). In its broad scientific meaning, biometrics concerns methods for detecting and measuring characteristics of living organisms and using those measurements for comparative analysis and classification.

In Computer Science, and specifically in this course, the term is used in the more focused sense of automated *identification* or *verification* of a person's identity from physical and/or behavioural characteristics. The course therefore aims to provide the basic knowledge and practical skills needed to design and develop automated systems for recognizing people from biometric features.

Key Concept

The course is not about biological measurement in general. Its central topic is automatic person recognition from biometric traits, including the complete pipeline from acquisition and representation to matching, decision, evaluation, and reliability.

Course Organization and Assessment

The source deck indicates the following assessment components:

- exercises during the course;
- a project involving the design and implementation of a biometric module using MATLAB or OpenCV, either individually or in a group (group work is strongly suggested in the source material);
- a classroom presentation, written report/essay, demonstration, and questions;
- Italian or English may be used for the assessment.

Reference Material Announced for the Course

The introductory deck points students first to the course slides and then to three general references that are also included in the bibliography of these notes:

- Jain, Flynn, and Ross, *Handbook of Biometrics*;
- Wechsler, *Reliable Face Recognition Methods*;
- Ross, Nandakumar, and Jain, *Handbook of Multibiometrics*.

Remark – Time-sensitive information

The source slides explicitly direct students to the updated course page for the current lesson schedule, classroom, and meeting time. These notes preserve that instruction instead of hard-coding potentially obsolete logistical details.

1 Introduction to Biometric Systems

Biometric recognition identifies or authenticates people from measurable physical or behavioural characteristics. The central engineering problem is to transform variable sensor observations into representations that remain discriminative across people and sufficiently stable for repeated use.

1.1 Biometrics as Pattern Recognition

A biometric system can be understood as a particular form of *pattern-recognition system*. A sample acquired from a person is represented through a collection of measurable characteristics, often summarized by a feature vector.

Two patterns are considered similar when an appropriate distance between their feature representations is sufficiently small, or equivalently when a similarity score is sufficiently large.

Three design questions determine whether a pattern-recognition pipeline can support reliable biometric matching:

1. Which distance or similarity measure is appropriate?
2. Which features are sufficiently informative and discriminative?
3. Which difference margin, or acceptance threshold, should be used to decide whether two patterns match?

Pattern recognition appears in different forms depending on the meaning of the classes.

In content-based image retrieval, classes can correspond to different object types, so the pattern representation separates, for example, a face from a flower.

In a more specific recognition problem, classes may correspond to subclasses of the same general type.

In biometrics, the classes are the *individuals themselves*: the pattern must distinguish one person from another.

Key Concept

Biometrics uses pattern-recognition techniques in a setting where the identities of individual people are the classes to be discriminated.

1.2 What Is Biometrics?

The term *biometrics* derives from the Greek words *bios* (life) and *métron* (measure).

In a broad sense, it refers to the measurement and statistical analysis of biological data.

In technological applications, biometrics concerns the measurement and analysis of physical and/or behavioural characteristics for recognizing or authenticating a person.

Definition – Biometrics

Biometrics is the automatic recognition of a person according to discriminative characteristics. In operational systems, measurable physical or behavioural traits are acquired, represented, compared, and used to support an identity decision.

The basic assumption is that each person is unique.

Turning this assumption into a working system requires three main steps:

- determine features that are sufficiently distinctive to identify a person;

- find reliable techniques for measuring those features;
- design reliable algorithms that can recognize or classify a person from the measured features.

1.3 Access Types

Biometric recognition can protect both physical and logical resources.

Table 1.1: Access types considered in biometric applications.

Access type	Examples
Physical access	Room, building, controlled area.
Logical access	Electronic resources, critical data.

1.4 Why Biometric Systems?

Traditional authentication commonly relies on two factors.

The first is *something one owns*, such as a card or document.

Such an object can be lost, stolen, or copied, and the system may effectively authenticate the object rather than the legitimate owner.

The second is *something one knows*, such as a password.

A password may be guessed, elicited from the user, or forgotten.

Moreover, a password that is easy to remember can also be easier to guess.

Biometric recognition adds a third possibility: *something one is*.

The identity evidence is derived from the person's own physical or behavioural characteristics.

Table 1.2: Authentication factors used in biometric authentication.

Factor	Example	Main limitation
Something one owns	Card, document	Can be lost, stolen, or copied; the object rather than its owner may be authenticated.
Something one knows	Password	Can be guessed, extracted from the user, or forgotten.
Something one is	Biometric trait	Recognition is based on measurable characteristics of the person; accuracy and trait-specific limitations remain.

1.5 Historical Background

Biometric recognition predates digital sensors and machine learning. Early identification systems already pursued the same two properties that remain central today: measurements should distinguish different people and should remain sufficiently stable over time.

1.5.1 Bertillon and Bertillonage

In 1882, Alphonse Bertillon (1853–1914), chief of the identification service of the Paris police, introduced an anthropometric identification procedure later known as *Bertillonage*. The method recorded bodily measurements and descriptive characteristics such as hand shape, bust and limb measurements, head shape, facial details, and the corresponding identification card.

The method was not based on one measurement alone. Measurements were divided into discrete categories so that records could first be partitioned into smaller groups and only then compared

in detail. The historical scheme described 243 primary categories; further subdivision using seven eye categories together with hair colour produced 1,701 groupings. With a collection of 5,000 records, a primary category would contain roughly 20 cards, reducing a search from the whole archive to a small candidate set.

Example – The Will West case

In 1903, a newly admitted prisoner named Will West was measured using the Bertillon procedure. The resulting measurements led to an existing record for William West, whose measurements and appearance were strikingly similar. Further investigation established that William West was already incarcerated and that the two men were different individuals. The case illustrates the central limitation of coarse anthropometric categorization: two people can fall into nearly the same measurement classes even when their identities are different.

The rapid growth of early identification archives also made indexing an engineering concern. Anthropometry was introduced into a United States prison setting in 1887 by Major McClaughry. A National Bureau of Identification was established in 1896 and was described as a forerunner of later federal identification services. The historical account reports 16,000 Bertillon cards after one year, 24,000 after two years, and 131 prisoners recorded as first offenders who were found to have prior records. By 1900, the prison administration was formally permitted to accept Bertillon cards.

Remark – Historical context

Bertillonage is important as an early model of biometric identification, not as a modern biometric specification. Its lasting contribution is the attempt to formalize human identification through measurable, persistent, and discriminative characteristics.

1.5.2 Galton and Fingerprints

At the end of the nineteenth century, Francis Galton criticized anthropometric identification from a statistical point of view. In 1892 he introduced the notion of a fingerprint *minutia* and proposed an early fingerprint classification method.

In 1893, the UK Home Office recognized the individuality of fingerprints. Police organizations subsequently began acquiring and storing fingerprint records and developing systematic methods for visual comparison. Around 1900, the Galton–Henry classification system became an important basis for fingerprint-recognition practice in many police organizations.

1.6 Limitations and Integration with Traditional Systems

Biometric recognition does not always work perfectly.

Recognition failures motivate integration with traditional authentication factors.

Combining different forms of evidence can increase the security level compared with relying on only one factor.

Figure 1.1 summarizes the three forms of authentication evidence discussed above.

1.7 Architecture of a Biometric System

A biometric system operates in two conceptually distinct phases: *enrollment* and *recognition*.

During enrollment, biometric data are captured and processed so that a reference template can be stored for future operations.

During recognition, newly acquired biometric data are converted into a probe template and compared with stored templates to support a decision.

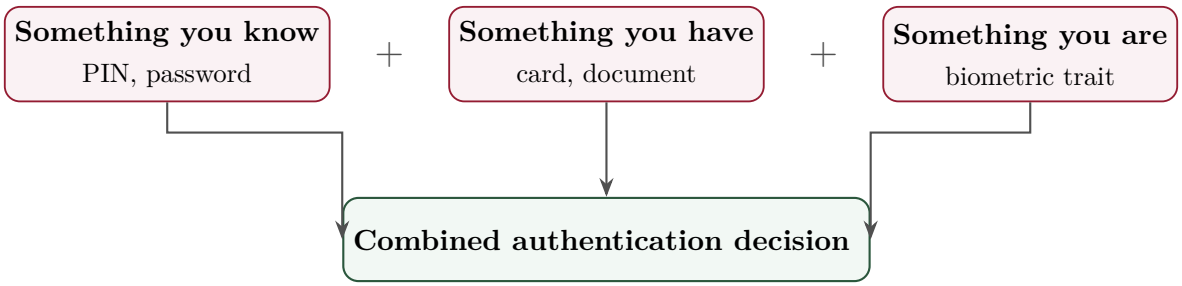


Figure 1.1: Combination of knowledge, possession, and biometric evidence.

Definition – Enrollment

Enrollment is the capture and processing of a user's biometric data for use by the system in subsequent authentication or recognition operations. The resulting reference templates populate the gallery.

Definition – Recognition

Recognition is the capture and processing of biometric data in order to produce an authentication or identity decision from the result of matching a current template against stored template information.

Definition – Probe and Gallery

A *probe* is a template submitted to the system for recognition. The *gallery* is the set of templates associated with enrolled subjects.

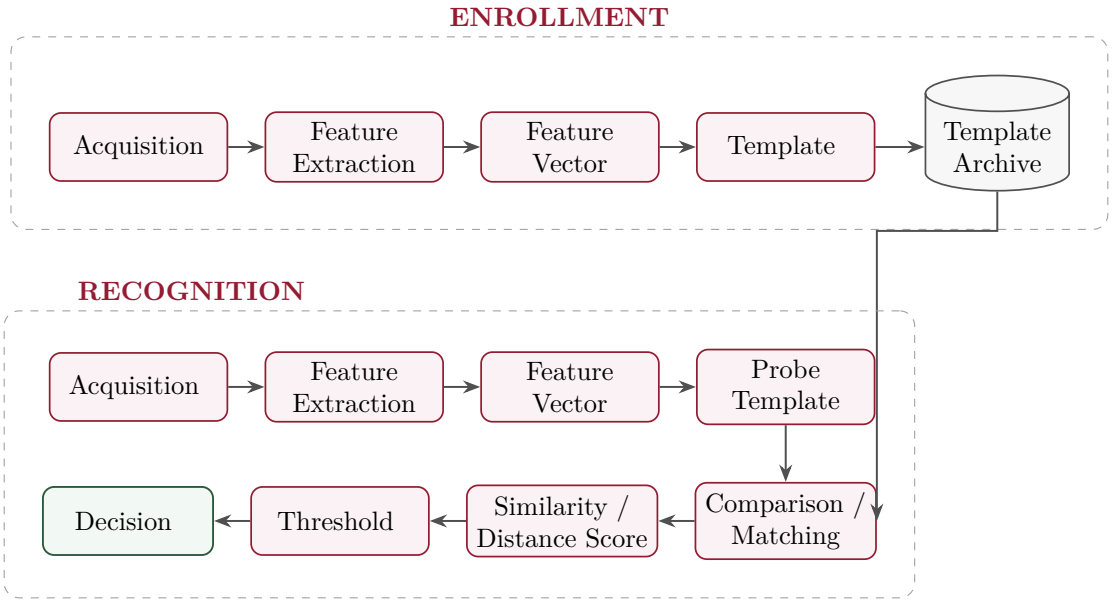


Figure 1.2: Conceptual architecture of enrollment and recognition in a biometric system.

Figure 1.2 separates the enrollment pipeline from the recognition pipeline and shows the stored template archive feeding the matcher.

Verification and identification use the same broad architecture but differ in the search space.

Verification performs a 1:1 comparison with the template(s) associated with a claimed identity.

Identification performs a 1:N search over the gallery because no identity is claimed in advance.

1.8 Modules of a Biometric System

A biometric system can be organized into four functional modules.

Table 1.3: Main modules of a biometric system.

Module	Role	Typical output
Sensor module	Captures the biometric data from the subject.	Raw sample, such as an image, recording, or other sensed signal.
Feature extraction module	Extracts the main characteristics from the acquired data. During enrollment it produces information used to form stored templates.	Feature vector or structured feature representation.
Matching module	Compares extracted features with stored templates.	One or more matching scores.
Decision module	Interprets matching results according to the operating rule.	Acceptance, rejection, or an identity result depending on the task.

1.9 Types of Users

Users can be classified along several dimensions.

Table 1.4: Operational categories of biometric-system users.

Category	Meaning
Cooperative / non-cooperative	A cooperative user is interested in being recognized. A non-cooperative user may be indifferent or actively adverse to recognition.
Public / private	Users may be customers/public users or employees/private users of the entity deploying the system.
Used / non-used	Describes how frequently the user interacts with the system: multiple times per day, daily, weekly, monthly, or only occasionally.
Aware / not aware	Describes whether the user is aware that a biometric recognition process is taking place.

1.10 Types of Acquisition Settings

In a *controlled* setting, capture conditions can be managed.

For face acquisition, for example, pose, illumination, and expression can be constrained.

Distortions can largely be avoided, defective templates can be rejected, and acquisition can be repeated.

In an *uncontrolled* or *undercontrolled* setting, capture conditions cannot be fully controlled.

Templates may exhibit different levels of distortion.

A defective template may still be detected and rejected, but the system may not have the opportunity to repeat the acquisition.

Table 1.5: Controlled and uncontrolled acquisition settings.

Setting	Capture characteristics	Consequence
Controlled	Pose, illumination, expression, and other conditions can be managed.	Distortion can often be reduced and acquisition may be repeated.
Uncontrolled / undercontrolled	Capture conditions cannot be fully managed.	Distortion may remain; defective templates can be detected but capture may not be repeatable.

1.11 Types of Recognition Operations

1.11.1 Verification

In verification, the user claims an identity, possibly by presenting an ID card or another identifier.

The system performs a 1:1 matching operation against the reference information associated with that identity.

The output is typically *accept* or *deny*.

1.11.2 Identification

In identification, the user makes no identity claim.

The system must determine whether the probe corresponds to one of the subjects in the gallery, which requires a 1:N operation.

The output is an identity when a suitable correspondence is found, with an explicit rejection option in open-set scenarios.

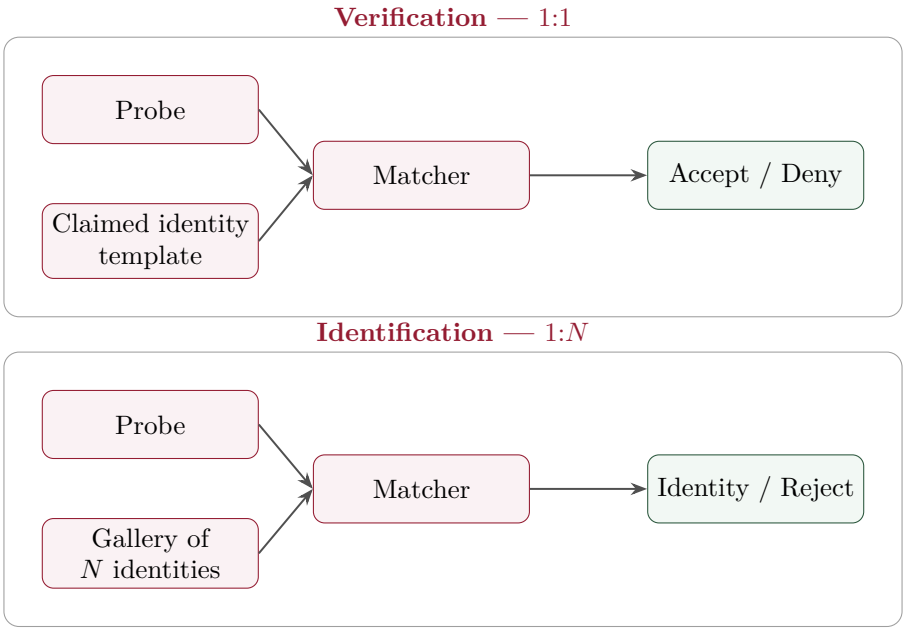


Figure 1.3: Verification compares a probe with a claimed identity, whereas identification searches a gallery.

The structural difference between the two operations is summarized in Figure 1.3.

1.11.3 Open-set, Closed-set, and Lists

In *open-set identification*, the system determines whether a probe p_i belongs to a subject represented in gallery G.

Some probes may not correspond to any enrolled subject, so the system needs a reject option.

Errors include rejecting an enrolled subject, accepting a non-enrolled subject, or returning the wrong enrolled identity.

In *closed-set identification*, all probes belong to enrolled subjects.

The principal error is returning the wrong identity.

A *watch list* contains subjects for whom the system checks membership.

A *white list* contains subjects who are granted access, whereas a *black list* contains subjects who are rejected or may trigger an alarm.

1.12 Requirements for a Biometric Trait

Table 1.6: Requirements for a biometric trait.

Requirement	Meaning
Universality	The trait should be possessed by every person, except for rare exceptions.
Uniqueness	Different people should be distinguishable according to the trait.
Permanence	The trait should remain sufficiently stable over time.
Collectability	The trait should be measurable by an appropriate sensor.
Acceptability	People should not object to collection or measurement of the trait.

The X9.84-2003 standard context includes fingerprints, iris and retina, face, ear, hand or finger geometry, signature, keystroke behaviour, voice, and DNA or other biological traces.

These modalities may be associated with physiological, behavioural, or mixed features.

1.13 Strong and Soft Biometric Traits

Examples of *strong biometric traits* include iris, face, and fingerprint, whereas *soft biometric traits* include hair colour, facial shape, and gait.

Soft traits may lack uniqueness or long-term persistence; behavioural characteristics can also be affected by transient conditions such as mood or health. Even so, soft traits can reduce the search space by narrowing candidate identities.

1.14 Examples of Biometric Modalities

A biometric modality determines both what is sensed and how the resulting signal should be represented. Different modalities therefore require different acquisition procedures, features, and matching strategies.

1.14.1 Voice

Voice recognition can represent speech through acoustic feature vectors. A Gaussian Mixture Model (GMM) can model the distribution of those features: training observations are collected for several acoustic dimensions, their statistical distributions are estimated, and the resulting model is used to score a new utterance. A representative score is a log-likelihood of the form

$$\text{score} = \log p(\text{speech} \mid \text{speaker model}).$$

A larger likelihood indicates that the observed acoustic vectors are more compatible with the stored model.

1.14.2 Signature

A signature may be treated as a static image or as a dynamic behavioural signal. Dynamic acquisition can retain the trajectory of the pen together with time-varying measurements. Representative features include the x- and y-coordinates of the pen, pressure, pen azimuth in 0°–359°, and pen altitude in 0°–90°. The resulting template can therefore contain substantially more information than the final ink trace alone.

1.14.3 Fingerprint

Fingerprint recognition uses information at different spatial levels. First-level information captures global ridge-flow structure and broad fingerprint patterns. Second-level information focuses on local minutiae, such as ridge endings and bifurcations.

A typical processing chain begins from the fingerprint image, estimates the orientation field, identifies a region of interest, extracts ridges, thins the ridge representation, and finally extracts minutiae. Global ridge patterns include arches, tented arches, loops, and whorls. Matching then requires comparing the resulting local structures rather than merely comparing two raw images pixel by pixel.

1.14.4 Iris

Iris recognition first captures an eye image and localizes the iris region. One representative pipeline begins from a 480×640 eye image. The annular iris texture is then normalized or *unwrapped*, encoded into a compact feature representation, and compared with enrolled templates. The comparison score is finally converted into an accept/reject decision. This pipeline separates geometric normalization from texture encoding so that matching can remain meaningful despite moderate changes in image scale or eye position.

1.14.5 Retina

Retinal recognition is based on the vascular pattern visible on the eyeground. The acquisition process captures the retina, localizes relevant vascular structures, derives a representation of the vessel pattern, and matches it against a stored template. The discriminative information is therefore associated with the geometry of the capillary network rather than with the external appearance of the eye.

1.14.6 Face

Face recognition admits several families of representations. Image-based approaches include methods based on Independent Component Analysis (ICA), neural networks, and eigenface-style representations. Feature-based approaches can compare structured facial landmarks, for example through elastic graph matching. Three-dimensional approaches can use 3D morphable models, while hybrid approaches combine multiple descriptions such as wavelet- or fractal-derived information.

Table 1.7: Representative biometric modalities and the information used for matching.

Modality	Representative information	Typical comparison idea
Voice	Acoustic feature vectors and their statistical distribution	Likelihood or vector-based comparison.
Signature	Static shape or dynamic pen trajectory, pressure, azimuth, altitude	Image, sequence, or feature-vector matching.
Fingerprint	Global ridge flow and local minutiae	Structured correspondence between ridge/minutia features.
Iris	Normalized iris texture encoding	Template comparison after localization and normalization.
Retina	Vascular pattern on the eyeground	Comparison of extracted vessel structures.
Face	Appearance, landmarks, learned features, or 3D structure	Image-, feature-, embedding-, or model-based comparison.

1.15 Benefits and Drawbacks of Biometric Traits

Biometric traits provide a natural form of authentication because the user need only appear in person.

They cannot be forgotten or lent in the same way as a password or token.

These advantages are accompanied by several drawbacks:

- biometric systems do not provide 100% accuracy;
- some users cannot be reliably recognized by some technologies, for example when fingerprints are heavily worn or damaged;
- traits may change over time;
- biometric devices may be unreliable under some circumstances;
- a copied biometric trait cannot simply be replaced in the same way as a compromised password.

Remark

A biometric characteristic is physically bound to the person, but a digital or physical representation of that characteristic can still be copied or compromised. Unlike a password, the underlying trait cannot simply be replaced after compromise.

1.16 Chapter Summary

Biometric systems treat personal identity as a pattern-recognition problem. A complete system must acquire a suitable trait, extract a stable representation, compare that representation with enrolled templates, and convert the resulting score into a decision. Verification performs a 1:1 comparison for a claimed identity, whereas identification searches a 1:N gallery and may additionally require a reject option. The practical quality of a modality depends on universality, uniqueness, permanence, collectability, and acceptability, while real deployments must also account for acquisition conditions, user cooperation, imperfect recognition, and the consequences of biometric compromise.

2 Performance Evaluation of Biometric Systems

A biometric matcher does not return certainty; it returns evidence, usually as a similarity or distance score. Performance evaluation studies how those scores behave for genuine and impostor comparisons and how an operating threshold converts them into correct decisions and errors.

2.1 Sources of Recognition Difficulty

Biometric recognition is difficult because the same person does not always produce identical samples, while different people can sometimes appear very similar according to the measured trait.

- **Large intra-class variation**: samples from the same person may vary substantially, for example because of pose or facial expression.
- **Small inter-class variation**: samples from different people may be very similar, as illustrated by twins or close relatives.
- **Noisy or distorted acquisitions**: poor fingerprint quality or non-uniform illumination can degrade the sample.
- **Non-universality**: some people may not provide a usable sample for a particular modality. About 4% of the population is reported to present poor-quality fingerprints, with the issue being more frequent in some groups such as elderly people.
- **Spoofing and attacks**: attacks may target different points of the biometric processing pipeline.

Potential attack points span the complete biometric pipeline: the sensor, feature extractor, communication channels, matcher, stored templates, and final decision path.

Examples include presenting a fake biometric, replaying old data, overriding feature extraction, synthesizing feature vectors, modifying stored templates, overriding the matcher, intercepting channels, or overriding the final decision.

Key Concept

Performance is not determined only by the matcher.
It depends on acquisition quality, representation quality, population variability, the operating threshold, and the integrity of the complete biometric pipeline.

2.2 Samples, Hand-crafted Features, and Templates

Definition – Sample

A sample is the raw captured biometric data, such as an image, a voice recording, or a fingerprint acquisition.

Definition – Hand-crafted Features

Hand-crafted features are characteristics manually designed by a data scientist and extracted from raw samples.

Definition – Template

A template is a collection of features extracted from the raw biometric data and used for comparison.

Templates may have different mathematical structures.

Representative template structures include:

- a histogram of relevant image values such as grey levels;
- a vector of measurements, such as a collection of anthropometric measurements;
- a time series, such as acceleration values recorded along the three accelerometer axes;
- a set of fingerprint minutiae represented by triplets

$$\{(x_1, y_1, \theta_1), \ldots, (x_n, y_n, \theta_n)\},$$

where the coordinates identify relevant points and θ_i represents the local ridge-tangent direction.

2.3 Comparing Templates

The comparison procedure depends on the representation.

Vector templates can often be compared with standard distances or similarities, whereas structured representations may require specialized matching.

2.3.1 Euclidean Distance

For vectors $\mathbf{p}, \mathbf{q} \in \mathbb{R}^n$, the Euclidean distance is

$$d(\mathbf{p}, \mathbf{q}) = \|\mathbf{p} - \mathbf{q}\|_2 = \sqrt{\sum_{i=1}^{n} (p_i - q_i)^2}. \tag{2.1}$$

Here n is the feature-space dimension, p_i and q_i are corresponding vector components, and $\|\cdot\|_2$ denotes the Euclidean norm. A smaller Euclidean distance indicates greater similarity.

2.3.2 Cosine Similarity

The cosine similarity between vectors $\mathbf{A}$ and $\mathbf{B}$ is

$$S_C(\mathbf{A}, \mathbf{B}) = \cos(\theta) = \frac{\mathbf{A}^\mathsf{T}\mathbf{B}}{\|\mathbf{A}\|_2 \, \|\mathbf{B}\|_2} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}. \tag{2.2}$$

Here θ is the angle between the two vectors, $\mathbf{A}^\mathsf{T}\mathbf{B}$ is their dot product, and $\|\mathbf{A}\|_2$, $\|\mathbf{B}\|_2$ are their Euclidean norms. Unlike a distance, larger cosine similarity values represent greater similarity.

2.3.3 Pearson Correlation

Pearson correlation can be used as a similarity measure for histograms or sets of paired values.

Given paired samples (x_i, y_i), $i = 1, \ldots, n$,

$$r_{xy} = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}. \tag{2.3}$$

An algebraically equivalent form is

$$r_{xy} = \frac{n\sum_i x_i y_i - \left(\sum_i x_i\right)\left(\sum_i y_i\right)}{\sqrt{n\sum_i x_i^2 - \left(\sum_i x_i\right)^2}\sqrt{n\sum_i y_i^2 - \left(\sum_i y_i\right)^2}}. \tag{2.4}$$

Here $\bar{x}$ and $\bar{y}$ are the sample means and n is the number of paired observations. The coefficient r_{xy} lies in the usual correlation range and is interpreted here as a similarity measure.

2.3.4 Bhattacharyya Distance

For discrete probability distributions P and Q over the same domain $\mathcal{X}$, the Bhattacharyya coefficient is

$$BC(P,Q) = \sum_{x\in\mathcal{X}} \sqrt{P(x)Q(x)}. \tag{2.5}$$

The corresponding distance is

$$D_B(P,Q) = -\ln\left(BC(P,Q)\right). \tag{2.6}$$

Here $P(x)$ and $Q(x)$ are the probability masses assigned to $x \in \mathcal{X}$, $BC(P,Q)$ is the Bhattacharyya coefficient, and $D_B(P,Q)$ is the corresponding distance. This measure is useful for comparing histogram-like or probabilistic representations.

2.3.5 Dynamic Time Warping

Dynamic Time Warping (DTW) is presented as a distance method for time series.

The motivating example compares two repetitions of a walking sequence recorded with motion capture.

Even when walking speeds differ, spatial trajectories may remain highly similar.

DTW allows a non-linear alignment of time indices before comparison.

2.3.6 Learned Features and Embeddings

In deep-learning systems, the final classification layer, typically a softmax layer, can be omitted when the goal is to extract the internal *embedding* that would otherwise feed the classifier.

These embeddings can then be treated as vectors and compared using vector-based similarity or distance measures.

Some representations require more than direct vector comparison.

Fingerprint minutiae, for example, require finding a suitable pairing between points before the match can be evaluated.

2.4 Similarity, Distance, and Acceptance Thresholds

Once a similarity or distance value has been obtained, it is compared with an acceptance threshold in verification and open-set identification.

For a similarity score s and threshold t,

$$\text{accept} \iff s \geq t. \tag{2.7}$$

For a distance d,

$$\text{accept} \iff d \leq t. \tag{2.8}$$

Reasoning with similarities and distances is symmetric after reversing the inequality and the interpretation of larger and smaller values.

Table 2.1: Comparison methods for representative biometric template structures.

Method	Type	Typical use
Euclidean distance	Distance	Vector templates.
Cosine similarity	Similarity	Vector templates and embeddings.
Pearson correlation	Similarity	Histograms or paired point sets.
Bhattacharyya distance	Distance	Histogram or distribution comparison.
Dynamic Time Warping	Distance	Time-series comparison with temporal misalignment.
Structured matching	Task-specific	Feature sets such as fingerprint minutiae where pairing is required.

2.5 Verification Outcomes

A verification claim is accepted when the best matching score for the claimed identity satisfies the threshold rule.

Four cases arise.

Table 2.2: Verification outcomes and common biometric terminology.

True situation	System decision	Biometric term	Alternative term
Genuine claim	Accept	Genuine Acceptance (GA)	Genuine Match (GM)
Genuine claim	Reject	False Rejection (FR)	False Non-Match (FNM)
Impostor claim	Reject	Genuine Rejection (GR)	Genuine Non-Match (GNM)
Impostor claim	Accept	False Acceptance (FA)	False Match (FM)

Remark – Terminology note

Type I and Type II depend on the choice of the null hypothesis. With H_0 = different person, as defined below, false acceptance is a Type I error and false rejection is a Type II error. The opposite labels require H_0 = same person.
The biometric names FAR and FRR are the less ambiguous terms to use when interpreting the hypothesis notation below.

Remark – Uncertain cases

A practical possibility between automatic acceptance and rejection is to route an uncertain score region to a human operator. Such a three-way policy is application-specific and does not change the definitions of GA, GR, FA, and FR used for binary evaluation.

2.6 False Acceptance Rate and False Rejection Rate

Performance Metric – False Acceptance Rate (FAR/FMR)

The False Acceptance Rate (FAR), also called False Match Rate (FMR), is the fraction or probability of impostor recognition operations in which the impostor is incorrectly accepted.

For example,

$$\text{FAR} = 0.1\%$$

means that, on average, one out of every 1000 impostor attempts is successful.

Performance Metric – False Rejection Rate (FRR/FNMR)

The False Rejection Rate (FRR), also called False Non-Match Rate (FNMR), is the fraction or probability of genuine recognition operations in which the genuine user is incorrectly rejected.

For example,

$$\text{FRR} = 0.05\%$$

means that, on average, one out of every 2000 authorized access attempts is incorrectly rejected.

2.7 Ground Truth and Evaluation Notation

Evaluation experiments use labelled samples, so the true identity is known through a ground-truth function $\text{id}(\cdot)$.

Notation	Meaning
$\text{id}(p_j)$	True identity of probe p_j.
$\text{id}(g_x)$	Identity associated with gallery template g_x.
$\text{topMatch}(p_j, i)$	Best match between probe p_j and the gallery template(s) associated with claimed identity i.
$s(t_1, t_2)$	Matching score between two templates.

Let $G = \{g_i\}$ denote the gallery templates. Let P_G be the set of probes belonging to subjects represented in the gallery and P_N the set of probes belonging to subjects not represented in the gallery.

For a genuine claim,

$$g_x = \text{topMatch}\left(p_j, \text{id}(p_j)\right), \qquad s_{xj} = s(g_x, p_j), \qquad p_j \in P_G. \tag{2.9}$$

For an impostor claim,

$$g_x = \text{topMatch}(p_j, i), \qquad i \neq \text{id}(p_j), \qquad s_{xj} = s(g_x, p_j). \tag{2.10}$$

Using the similarity-threshold convention,

$$\text{FRR}(t) = \frac{|\{p_j : s_{xj} < t,\ \text{id}(g_x) = \text{id}(p_j)\}|}{|\{p_j : \text{id}(g_x) = \text{id}(p_j)\}|}. \tag{2.11}$$

Similarly,

$$\text{FAR}(t) = \frac{|\{p_j : s_{xj} \geq t,\ \text{id}(g_x) \neq \text{id}(p_j)\}|}{|\{p_j : \text{id}(g_x) \neq \text{id}(p_j)\}|}. \tag{2.12}$$

Two impostor scenarios are possible. An impostor may itself be registered but claim another identity, or may be completely outside the gallery.

For verification error computation, both are false claims.

2.8 Genuine and Impostor Score Distributions

A score is *genuine* when it results from matching two samples from the same enrolled individual.

A score is *impostor* when it results from matching samples associated with different identities.

The decision problem can be written through the hypotheses

$$H_0 : \text{different person},$$
$$H_1 : \text{same person}.$$

Possible decisions are

$$D_0 : \text{decide different person},$$
$$D_1 : \text{decide same person}.$$

The corresponding probabilities are

$$\text{FAR} = \Pr(D_1 \mid H_0), \qquad \text{FRR} = \Pr(D_0 \mid H_1). \tag{2.13}$$

Figure 2.1 gives a schematic view of the two score distributions and the effect of a similarity threshold.

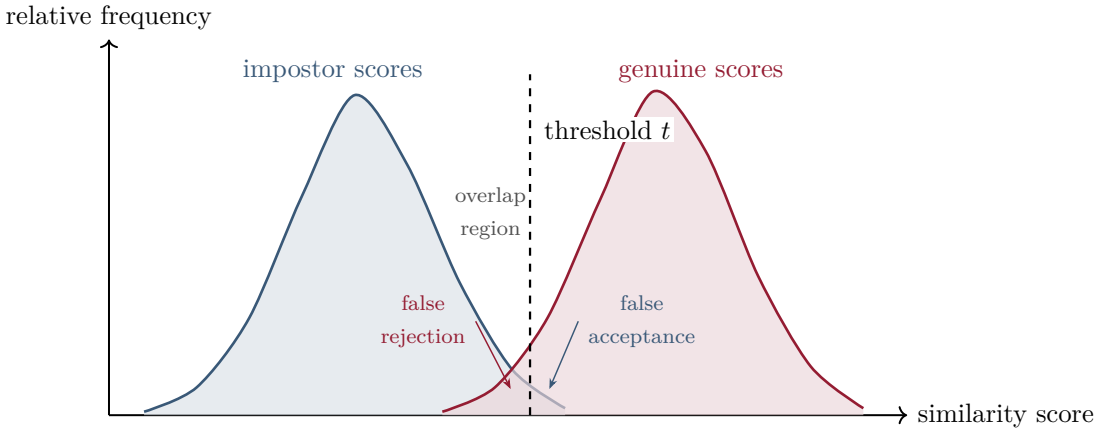


Figure 2.1: Schematic genuine and impostor similarity-score distributions. Numerical coordinates are used only to draw the conceptual shapes; they do not represent experimental measurements.

2.9 GAR, EER, ZeroFAR, and ZeroFRR

The Genuine Acceptance Rate (GAR), also called Genuine Match Rate (GMR), is

$$\text{GAR} = 1 - \text{FRR}. \tag{2.14}$$

The Equal Error Rate (EER) is the error level obtained at a threshold t^* where

$$\text{FAR}(t^*) = \text{FRR}(t^*). \tag{2.15}$$

Two limiting operating points are also useful:

- **ZeroFRR / ZeroFNMR**: the FAR obtained when FRR is zero;
- **ZeroFAR / ZeroFMR**: the FRR obtained when FAR is zero.

For distance-based matching, an excessively low threshold produces many genuine rejections, while an excessively high threshold produces many false acceptances.

Under the similarity-threshold convention used in Equations (2.7), (2.11), and (2.12), the direction is reversed: increasing the threshold makes acceptance stricter, so FAR decreases while FRR increases.

The EER threshold is a convenient and common comparison point because it balances the two verification error rates. It is not automatically the best operating threshold for a deployed application: the operating threshold must follow the relative cost of false acceptances and false rejections.

Figure 2.2 shows the corresponding threshold trade-off using the same similarity-threshold convention as Equations (2.7), (2.11), and (2.12).

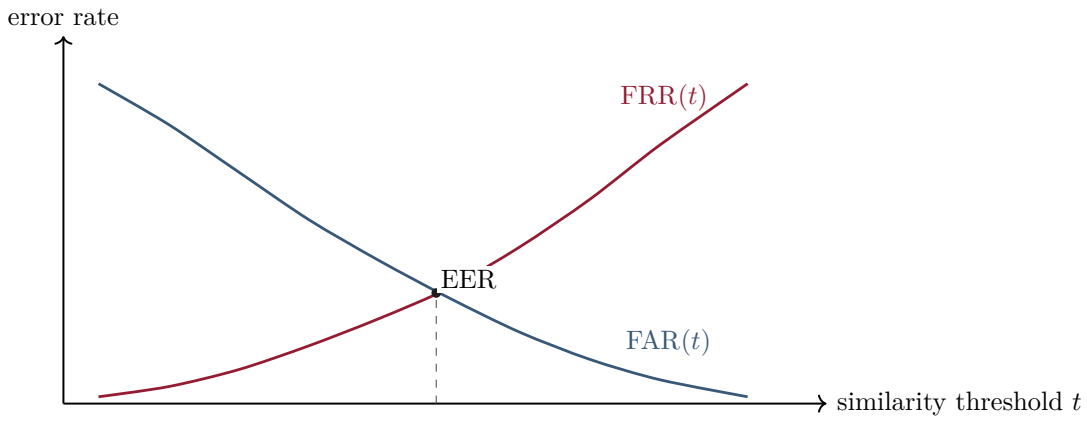


Figure 2.2: Schematic FAR/FRR trade-off under a *similarity-threshold* convention: increasing t makes acceptance stricter, so FAR decreases while FRR increases. The intersection illustrates EER; no experimental values are implied.

2.10 ROC and DET Curves

Performance Metric – ROC

The Receiver Operating Characteristic (ROC) plots Genuine Acceptance Rate

$$\mathrm{GAR} = 1 - \mathrm{FRR}$$

against False Acceptance Rate (FAR) as the threshold varies.

Performance Metric – DET

The Detection Error Trade-off (DET) curve plots False Rejection Rate (FRR) against False Acceptance Rate (FAR).
The standard DET uses normal-deviate (probit) axes: the plotted coordinates are $\Phi^{-1}(\mathrm{FAR})$ and $\Phi^{-1}(\mathrm{FRR})$, where Φ is the standard normal CDF. This is not a logarithmic scale.

Each operating threshold corresponds to a point on the relevant performance curve.

The two curve types are illustrated schematically in Figures 2.3 and 2.4.

Key Concept

For ROC curves, better systems tend toward high GAR at low FAR, i.e. toward the upper-left operating region. For DET curves, better systems tend toward simultaneously low FAR and low FRR, i.e. toward the lower-left region. Curve comparisons must still be made at operating points relevant to the intended application.

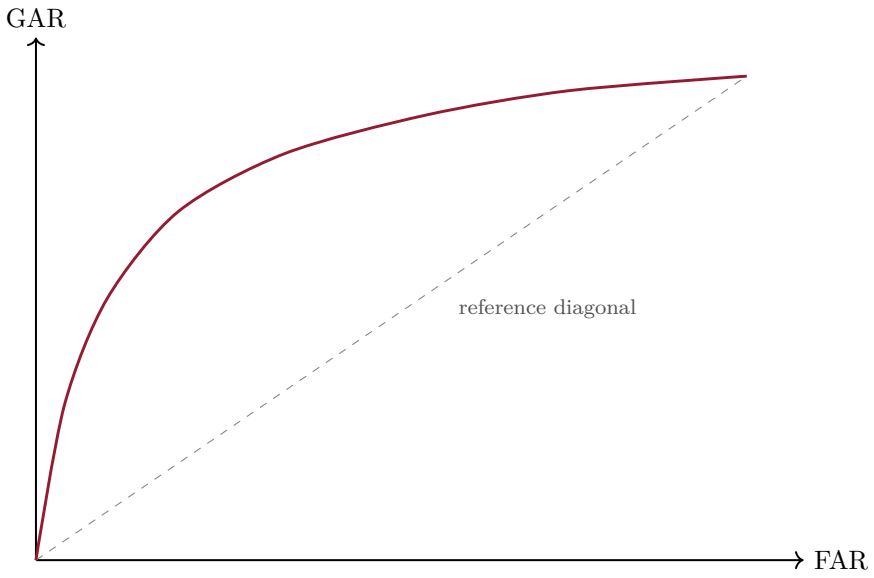


Figure 2.3: Schematic ROC representation with FAR on the horizontal axis and GAR on the vertical axis. The curve is illustrative only and contains no measured performance values.

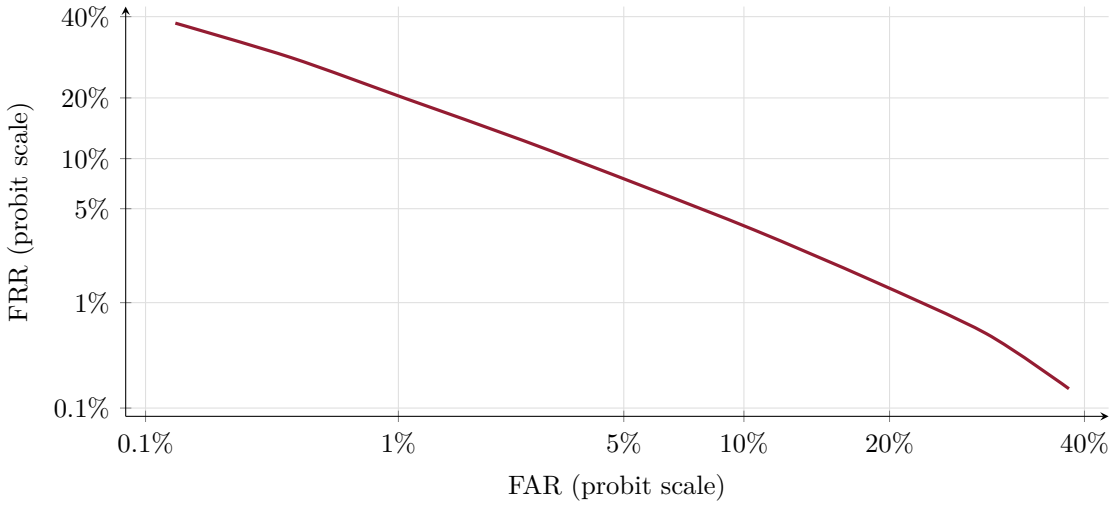


Figure 2.4: Schematic DET representation. Both axes are normal-deviate (probit) transforms of the corresponding error probabilities; the tick labels show the original FAR and FRR percentages. The curve is conceptual and does not encode experimental measurements.

2.11 Open-set Identification Performance

In open-set identification, such as a watch-list application, the system addresses two questions:

1. Is the probe subject represented in the gallery?
2. If so, who is the probe subject?

Open-set identification combines ranked similarity scores with an acceptance threshold.

For a probe belonging to a gallery subject, correct operation requires both detection above the threshold and correct ranking of the true identity.

For a probe outside the gallery, correct operation requires that no gallery candidate incorrectly triggers an alarm.

Typical open-set applications include comparing people observed in an airport against a watch list or comparing an unidentified person against a missing-person database. These applications require both a membership decision and, when a match is detected, an identity decision.

Example – Thresholded ranked scores

Suppose four candidates obtain scores

$$0.90,\ 0.86,\ 0.60,\ 0.40.$$

With a similarity threshold of 0.85, two candidates exceed the threshold.
If the top-ranked candidate is the correct identity, the operation is both a correct detection and a correct identification.
Raising the threshold to 0.95 produces no detection.
If the correct identity is not the highest-scoring candidate, detection can occur without correct identification. For example, with scores $0.86, 0.80, 0.60, 0.40$ and threshold 0.75, detection occurs, but the operation is not a correct detect-and-identify if the true identity is the second-ranked candidate.
For a probe from outside the gallery, scores $0.80, 0.70, 0.60, 0.40$ with threshold 0.85 produce no alarm and therefore a correct rejection. Lowering the threshold to 0.75 makes the highest score trigger an alarm, which is a false alarm because the probe has no mated gallery identity.

For probes from P_G, repeated trials estimate how often the system correctly detects and identifies the subject.

This quantity is the *correct detect and identify rate*.

For probes from P_N, repeated trials estimate how often an incorrect alarm occurs.

This is the *false alarm rate*.

2.11.1 Rank and DIR

Let $\mathrm{rank}(p_j)$ denote the position in the ranked list at which the first template for the correct identity appears.

The Detection and Identification Rate, $\mathrm{DIR}(t, k)$, expresses the probability that a gallery probe is detected above operating threshold t and that the correct identity appears within the first k ranked candidates. For similarity scores, a precise form is

$$\mathrm{DIR}(t,k) = \frac{\left|\left\{p_j \in P_G : \mathrm{rank}(p_j) \le k, \max_{g_i \in G:\, \mathrm{id}(g_i)=\mathrm{id}(p_j)} s(g_i, p_j) \ge t\right\}\right|}{|P_G|}. \tag{2.16}$$

Here k is the maximum accepted rank, t is the similarity threshold, and the inner maximum selects the best mated gallery template for probe p_j.

At rank 1, the corresponding False Negative Identification Rate (FNIR) is

$$\mathrm{FNIR}(t) = 1 - \mathrm{DIR}(t, 1). \tag{2.17}$$

The explicit threshold dependence is important in open-set evaluation because changing t changes both the detection probability and the false-alarm rate.

The false-alarm counterpart is the False Positive Identification Rate (FPIR), also described as FAR or false alarm rate in a watch-list context. For non-enrolled probes,

$$\mathrm{FPIR}(t) = \frac{|\{p_j \in P_N : \max_{g_i \in G} s(g_i, p_j) \geq t\}|}{|P_N|}. \tag{2.18}$$

Thus G is the gallery, P_N is the set of non-gallery probes, and an alarm occurs whenever at least one gallery score reaches the similarity threshold. The same quantity is also described using FAR/false-alarm terminology in some contexts; FPIR is the more task-specific name in open-set identification.

At rank 1, an open-set equal-error operating point can be described by $\mathrm{FNIR}(t) = \mathrm{FPIR}(t)$, i.e. by equalizing the paired open-set error probabilities.

2.11.2 Open-set ROC and Operating Regions

For a fixed rank limit, the detect-and-identify rate and false-alarm rate vary in the same direction with the similarity threshold: both are non-increasing as the threshold rises.

Raising a similarity threshold tends to reduce both successful detection and false alarms.

Lowering the threshold tends to increase both.

Five practical operating preferences can be distinguished:

1. extremely low false alarm;
2. extremely high detection and identification probability;
3. low false alarm with lower detection and identification;
4. high detection and identification with higher false alarms;
5. no fixed threshold, with ranked results returned for investigation.

2.12 Closed-set Identification Performance

Closed-set identification is a special case in which every probe is known to have a corresponding identity in the gallery.

The membership question is therefore already answered.

Pure closed-set operation is relatively restrictive in practice. A system may be described broadly as an identification system while operationally behaving like a watch list; the FBI Integrated Automated Fingerprint Identification System (IAFIS) is a representative example of this distinction.

The remaining issue is how highly the correct identity is ranked.

If the correct identity has the highest similarity score, the system has achieved correct identification at rank 1.

If the correct identity has the second-highest score, it is within the top two results and contributes to correct identification at rank 2.

Example – Closed-set ranks

With similarity scores $0.90, 0.86, 0.60, 0.40$, a correct identity in the first position gives a rank-1 identification. With scores $0.86, 0.80, 0.60, 0.40$, if the correct identity is the second-ranked candidate, the result contributes to identification at rank 2 even though it is not correct at rank 1. More generally, rank k asks whether the correct identity appears anywhere within the first k candidates.

Performance Metric – Cumulative Match Score (CMS)

The CMS at rank k is the probability that the correct identity is found among the first k positions of the ranked candidate list.
Equivalently, it can be estimated as the fraction of test probes whose correct identity appears within the top k results.

Performance Metric – Cumulative Match Characteristic (CMC)

A CMC curve plots the cumulative match score as a function of rank.
CMS at rank 1 gives the probability that the correct identity is the first returned candidate.
CMS at rank 2 gives the probability that it appears in either of the first two positions.
More generally, CMS at rank k gives the probability that the correct identity appears among the first k positions.

When the rank reaches the gallery size, the cumulative probability reaches 1 because the correct identity must occur somewhere in the complete closed-set list.

CMS at rank 1 is also called the *Recognition Rate (RR)*.

Figure 2.5 visualizes this cumulative ranking behaviour.

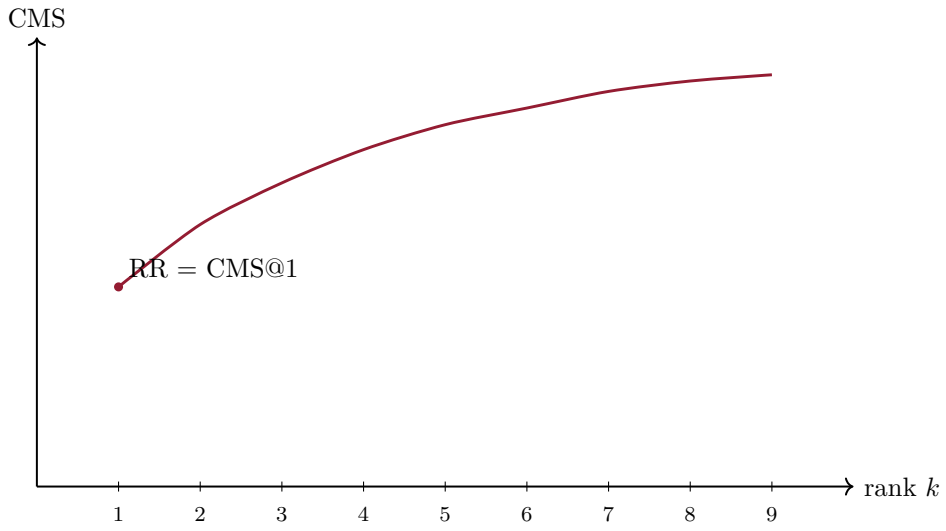


Figure 2.5: Schematic cumulative match characteristic (CMC) for closed-set identification. The curve illustrates cumulative ranking behaviour only; no dataset-specific values are implied.

2.13 Avoiding Confusion with Machine-Learning Metrics

Biometric error measures should not be confused with precision, recall, and F-score.

In standard machine-learning notation, let TP, FP, TN, and FN denote true positives, false positives, true negatives, and false negatives, respectively. Then

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}. \tag{2.19}$$

The F-score combines precision and recall; it is therefore also conceptually different from FAR and FRR.

In biometric acceptance/rejection terms, precision can be compared with

$$\frac{GA}{FA + GA},$$

while recall can be compared with

$$\frac{GA}{GA + FR}.$$

The biometric error rates of primary interest are instead

$$\text{FRR} = \frac{FR}{GA + FR}, \qquad \text{FAR} = \frac{FA}{FA + GR}. \tag{2.20}$$

Under the mapping acceptance $\leftrightarrow$ positive and rejection $\leftrightarrow$ negative,

$$\text{FRR} = \frac{FN}{TP + FN}, \qquad \text{FAR} = \frac{FP}{FP + TN}. \tag{2.21}$$

Thus FRR corresponds to the false-negative rate or miss rate, while FAR corresponds to the false-positive rate or fall-out.

Precision and recall therefore do not express the same error statistics as FAR and FRR.

2.14 Identification versus Re-identification

Biometric identification and re-identification (re-ID) address related but different tasks.

Table 2.3: Identification and re-identification as related recognition tasks.

Task	Objective	Time perspective / context
Identification	Determine the identity of an unknown subject.	Medium-to-long-term identity recognition.
Re-identification	Match the presence of the same person across different cameras, locations, video segments, image sequences, or event sets using appearance, body shape, behaviour, and tracking information.	Short-term association across non-overlapping time slices.

Re-identification is evaluated with metrics adapted to ranked retrieval.

CMC is appropriate when there is a single mated gallery image for each query.

When several mated images may exist, Average Precision (AP) and mean Average Precision (mAP) are used.

AP is described as the area under a precision–recall curve computed over thresholds, while mAP averages AP over queries.

Re-identification can connect back to identification when a person is repeatedly found in video and the resulting identity evidence is then checked against a watch list.

2.15 Chapter Summary

Performance evaluation connects matcher scores to operational error probabilities. The principal measures for verification and identification are consolidated below.

Table 2.4: Compact summary of central biometric performance measures.

Measure	Main task	Meaning
FAR / FMR	Verification	Probability or fraction of impostor claims incorrectly accepted.
FRR / FNMR	Verification	Probability or fraction of genuine claims incorrectly rejected.
GAR / GMR	Verification	Genuine acceptance probability; $1 - \text{FRR}$.
EER	Verification / open set	Operating point where paired error rates are equal.
DIR	Open-set identification	Detection and identification performance at a chosen rank.
FNIR	Open-set identification	False negative identification rate.
FPIR	Open-set identification	False positive identification or false alarm probability.
CMS@k	Closed-set identification	Probability that the correct identity appears among the first k ranked candidates.
CMC	Closed-set identification	Curve of CMS values across ranks.
RR	Closed-set identification	Recognition rate, equal to CMS at rank 1.

3 Recognition Reliability

Performance measures such as FAR, FRR, ROC, DIR, and CMC summarize the behaviour of a biometric system over a set of trials. Lesson 3 extends that view: a system can have acceptable global figures of merit while individual responses differ substantially in quality and reliability. The lecture therefore distinguishes *global performance*, *input quality*, and *response reliability*.

3.1 Why Global Performance Measures Are Not Enough

Identification is generally more difficult than verification, both for biometric systems and for human operators. It is therefore essential to reason about the correct task and its associated statistics rather than mixing verification and identification measures.

The lecture highlights two limits of relying only on FAR, FRR, CMS, and similar aggregate measures:

- performance is measured *ex post* after algorithms have been trained and tuned on finite data, so a benchmark can become a "closed world" that does not fully represent deployment conditions;
- the operational context can change, and a change in acquisition conditions can change the score distributions themselves.

For a meaningful comparison of systems, the source material asks us to consider not only a single performance number but also the number and characteristics of the databases, image size, probe and gallery sizes, the amount and quality of addressed/tolerated variation, and interoperability or cross-dataset generalization.

Remark – Dataset bias

The Torralba–Efros observation cited in the lecture is used as a warning: a benchmark is both a source of data and a measurement instrument. Optimizing only for the benchmark number can narrow the problem until the dataset behaves like a world of its own.

3.2 Distance Matrices and Error Computation

Given probes p_i and gallery templates g_j, the matcher can place each pairwise distance in a distance matrix

$$\mathrm{DM}_{ij} = d(p_i, g_j).$$

If an all-against-all experiment uses the same set on both axes and d is symmetric, then DM is symmetric. Self-comparisons produce a zero diagonal when the distance satisfies identity of indiscernibles.

For a distance-based matcher and threshold t, a false acceptance occurs when

$$\mathrm{DM}_{ij} \le t \quad \text{while} \quad \mathrm{id}(p_i) \neq \mathrm{id}(g_j),$$

whereas a false rejection occurs when

$$\mathrm{DM}_{ij} > t \quad \text{while} \quad \mathrm{id}(p_i) = \mathrm{id}(g_j).$$

These trial counts, repeated over threshold values, give the FAR/FRR curves introduced in Lesson 2. Here the acceptance rule is $d \le t$, so a distance equal to the threshold is accepted.

3.2.1 Metric, Semimetric, and Symmetrization

A metric on a set X is a function $d : X \times X \to \mathbb{R}$ such that, for all $x, y, z \in X$,

$$\begin{aligned} &d(x, y) \geq 0, \\ &d(x, y) = 0 \iff x = y, \\ &d(x, y) = d(y, x), \\ &d(x, z) \leq d(x, y) + d(y, z). \end{aligned}$$

The last property is the triangle inequality. In the terminology used by the lecture, a *semimetric* satisfies the first three properties but is not required to satisfy the triangle inequality.

If a comparison measure is not symmetric, the lecture proposes the simple symmetrization

$$d^*(x, y) = d^*(y, x) = \frac{d(x, y) + d(y, x)}{2}.$$

3.3 Relations Among Evaluation Measures

The lecture recalls the result by Bolle et al. that CMC and ROC-style performance are related when identification ranking is produced by repeatedly using a 1:1 matcher. Under that construction, the CMC is not an independent source of information: it is another representation of behaviour already linked to the matcher's FAR and FRR curves.

Starting from a distance or similarity matrix, many different performance quantities can be computed. The slides refer to these summary quantities as *Figures of Merit* (FoMs).

3.4 The Biometric Menagerie

Doddington et al. introduced a useful vocabulary for describing subject-specific matching behaviour. The terminology originated in speaker recognition but is used in the lecture as a general conceptual tool.

Table 3.1: Doddington's biometric menagerie as presented in the lecture.

Category	Characteristic behaviour
Sheep	Genuine comparisons score well and impostor comparisons score poorly; this is the normal, relatively easy behaviour with fewer errors than problematic users.
Goats	Genuine comparisons score poorly, so false rejections are more likely.
Lambs	The subject is comparatively easy to impersonate; impostor comparisons against this identity can score unusually high, increasing false matches.
Wolves	The subject is comparatively good at impersonating others and has an above-average chance of producing high impostor match scores.

Yager and Dunstone extend the menagerie using both a user's average genuine score G_k and average impostor score I_k:

- **Chameleons**: high G_k and high I_k; genuine matching is strong but impostor matching can also be high, so false accepts are a concern while false rejects are relatively uncommon.
- **Phantoms**: low G_k and low I_k; scores tend to be low regardless of partner, leading especially to false rejects.
- **Doves**: high G_k and low I_k; highly recognizable users that match themselves well and others poorly.
- **Worms**: low G_k and high I_k; the most problematic region, because genuine scores are weak while impostor scores can be strong.

Key Concept

Global accuracy can hide subject-specific behaviour. The menagerie is a way to reason about who creates errors and why, not merely how many errors the system makes in aggregate.

3.5 Input Quality and Response Reliability

A central distinction in Lesson 3 is that *sample quality* and *system-response reliability* are different concepts.

A quality measure evaluates the input before or independently of the final recognition response. A reliability measure instead asks whether a particular output of the biometric system should be trusted. Per-response reliability can support operational policies (for example, recapturing a sample when the result is not reliable enough) and can also support fusion in multibiometric systems.

3.5.1 Quality from Correlation and Sharpness

One family of approaches models a reference "average" image and assesses a sample by correlation with that reference. Another family estimates image sharpness: blur removes high-frequency detail, so sharpness provides evidence about acquisition quality.

3.5.2 Face-specific Quality Measures

The face-quality formulation attributed in the lecture to De Marsico, Nappi, and Riccio uses several components. With the notation of the source slides, pose distortion is expressed as

$$SP = \alpha(1 - \text{roll}) + \beta(1 - \text{yaw}) + \gamma(1 - \text{pitch}).$$

A grey-level homogeneity term is written as

$$SI = 1 - F(\text{std}(m_c)),$$

and a symmetry term as

$$SY = \sum_{(i,j)\in X} \text{sym}(P_i, P_j).$$

Here F, m_c, X, and sym are kept in the source-specific notation used by the lecture; the slides define the conceptual role of the terms rather than expanding those helper functions on the page.

3.5.3 Universal Image Quality Index

For original image samples $x = \{x_i\}_{i=1}^N$ and test-image samples $y = \{y_i\}_{i=1}^N$, the lecture gives the Universal Image Quality Index (UIQI)

$$Q = \frac{4\sigma_{xy}\,\bar{x}\,\bar{y}}{(\sigma_x^2 + \sigma_y^2)(\bar{x}^2 + \bar{y}^2)},$$

with

$$\bar{x} = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \bar{y} = \frac{1}{N}\sum_{i=1}^{N} y_i,$$

$$\sigma_x^2 = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})^2, \qquad \sigma_y^2 = \frac{1}{N-1}\sum_{i=1}^{N}(y_i - \bar{y})^2,$$

and

$$\sigma_{xy} = \frac{1}{N-1}\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y}).$$

The lecture interprets image distortion through correlation loss, luminance distortion, and contrast distortion.

3.5.4 Sharpness Estimation Quality Index

For an image of $x \times y$ pixels with intensity $p_{n,m}$, the source defines an average of horizontal and vertical adjacent-pixel differences:

$$\mathrm{SE} = \frac{1}{2}\left(\frac{1}{(x-1)y}\sum_{m=1}^{y}\sum_{n=1}^{x-1}|p_{n,m} - p_{n+1,m}| + \frac{1}{(y-1)x}\sum_{m=1}^{y-1}\sum_{n=1}^{x}|p_{n,m} - p_{n,m+1}|\right).$$

3.6 Evaluating a Quality Measure

The lecture uses FERET, LFW, and SCface as example test sets and proposes two levels of analysis:

1. inspect the distribution of quality values over a dataset and the correlation between different quality measures;
2. vary a tolerance threshold, discard samples judged too distorted, and measure the effect on recognition error (for example EER).

A useful quality measure should reduce recognition error while discarding as few samples as possible.

3.7 Margins Based on Error Estimation

The Poh–Bengio approach shown in the lecture evaluates errors as functions of a score-dependent quantity Δ and defines the margin

$$M(\Delta) = |\mathrm{FAR}(\Delta) - \mathrm{FRR}(\Delta)|\,.$$

The figure in the source illustrates how the margin changes with the score and therefore how error behaviour can be used as an indication of confidence.

3.8 System Response Reliability (SRR)

System Response Reliability (SRR) is introduced as a per-probe index in $[0, 1]$ for an identification response. The idea is to measure the amount of confusion around the identity returned by the system. A sparse neighbourhood around the first candidate suggests a more reliable response; a crowded neighbourhood suggests ambiguity.

Assume that the identification output is the gallery G ordered by distance from a probe p, and let g_{i_1} and g_{i_2} denote the first and second ranked candidates. The lecture studies two functions φ.

3.8.1 Relative-distance Function

The relative-distance form is

$$\varphi_{\mathrm{RD}}(p) = \frac{F(d(p, g_{i_2})) - F(d(p, g_{i_1}))}{F(d(p, g_{i_{|G|}}))}.$$

The numerator measures the separation between the best two candidates, while the denominator normalizes it by the largest distance-derived value in the ranked list. A small top-two separation relative to the global spread implies more confusion and lower reliability.

3.8.2 Density-ratio Function

The density-ratio form is

$$\varphi_{\mathrm{DR}}(p) = 1 - \frac{|N_b|}{|G|},$$

where the lecture defines

$$N_b = \{g_{i_k} \in G \mid F(d(p, g_{i_k})) < 2\, F(d(p, g_{i_1}))\}\,.$$

For the example in the source, both $|N_b|$ and $|G|$ are counted without the first-ranked item so that a maximum value of 1 remains possible. The slides note that this function is less sensitive to outliers than the relative-distance variant, while its fixed factor 2 makes the effective neighbourhood depend on the distance of the first candidate.

3.8.3 Critical Value and Normalized SRR

A critical value $\bar{\varphi}$ is selected to minimize reliability-estimation mistakes: false acceptances supported by a high φ and genuine cases not supported because φ is too low. The distance from the current value $\varphi(p)$ to $\bar{\varphi}$ is therefore used as reliability evidence.

The source defines the piecewise normalization

$$S(\varphi(p), \bar{\varphi}) = \begin{cases} 1 - \bar{\varphi}, & \varphi(p) > \bar{\varphi}, \\ \bar{\varphi}, & \text{otherwise}, \end{cases}$$

and then writes the normalized index as

$$\mathrm{SRR} = \frac{\varphi(p) - \bar{\varphi}}{S(\varphi(p), \bar{\varphi})}.$$

The final expression in the slide abbreviates the denominator notation, but the piecewise scale is the quantity defined immediately above it.

3.8.4 Adaptive Reliability Threshold

To derive an operational reliability threshold from a sequence of observations, the lecture seeks a high mean reliability together with low variability. In the notation of the source, the proposed threshold is

$$th_i = \left| \frac{\mathbb{E}[\bar{S}_i]^2 - \sigma[\bar{S}_i]}{\mathbb{E}[\bar{S}_i]} \right|.$$

3.9 Template Updating

Recognition operates on templates derived from biometric samples. Different captures of the same individual can generate different templates, and the system accumulates additional biometric data during operation. The lecture therefore presents template updating as a way to address *template ageing* and to enhance the gallery representation over time.

Two design choices are emphasized:

- **label assignment**: supervised systems use an external supervisor to label newly acquired data and typically operate offline; semi-supervised systems exploit both labeled and unlabeled data and may operate online or offline;
- **representative-template selection**: online selection is performed as new samples arrive, while offline selection is performed after a quantity of data has accumulated over a time interval.

3.10 Chapter Summary

Lesson 3 extends aggregate performance analysis to the reliability of individual responses. The central chain is

$$\text{input quality} \longrightarrow \text{score behaviour} \longrightarrow \text{response reliability} \longrightarrow \text{operational action.}$$

Dataset bias, subject-specific matching behaviour, quality screening, SRR, and template updating all address aspects that cannot be summarized by a single FAR, FRR, EER, or CMC value.

4 Face Recognition: Introduction and Face Localization

4.1 Why Face Biometrics?

The lecture frames biometric feasibility around two especially important factors: *accuracy/reliability* and *acceptability*. DNA is extremely accurate but requires intrusive acquisition procedures. Fingerprints can be accurate and widely accepted, but acquisition normally requires an aware and collaborating user and some users may present poor-quality fingerprints. Face biometrics occupies a different point in this design space: image capture is highly acceptable and can even occur without the user being aware of the capture, while recognition accuracy remains strongly dependent on conditions.

Further practical advantages listed in the lecture include the naturalness of recognizing people by face, the widespread social acceptability of photography, easy deployment of cameras, integration with logical access, and use in remote control or surveillance scenarios.

Key Concept

Face recognition is attractive not because the face is an easy biometric trait, but because acquisition is exceptionally convenient. The engineering challenge is to control or model the large variability of facial appearance.

4.2 The Face as a Complex Biometric Object

A face contains many potentially discriminative points and regions, and its appearance changes under pose, illumination, expression, occlusion, ageing, and camera/environment effects. The lecture uses MPEG-4 facial feature points to illustrate that a face is a structured object with many local components rather than a single rigid pattern.

The comparison charts in the source are intended to emphasize trade-offs among traits. Face shares several desirable practical properties with fingerprints, but there is no universally best biometric characteristic: the importance of acceptability, accuracy, hygiene, cost, universality, collectability, and resistance to circumvention depends on the application.

4.3 Applications and Recognition Challenges

Examples in the lecture include forensic recognition, logical/mobile access, border control, and recognition in a crowd. These applications differ in capture distance, cooperation, background clutter, pose, and the number of people in the scene.

Two fundamental statistical difficulties reappear in face recognition:

- **intra-personal variation**: the same person can look very different under changes of expression, illumination, pose, age, or appearance;
- **inter-personal similarity**: different people can have very similar facial appearance.

The lecture groups three major nuisance factors as **PIE**: Pose, Illumination, and Expression. Adding ageing produces the mnemonic **A-PIE**.

4.4 Face Databases and Controlled versus Uncontrolled Data

Public datasets are used both for training and for reproducible evaluation. The lecture distinguishes datasets with controlled, known variations from unconstrained data where the precise cause of

appearance variation may not be known.

4.4.1 FERET

FERET contains 14,051 images divided into multiple categories. The lecture emphasizes pose, illumination, and time variations. A gallery of 1,196 images and several probe groups (including fafb, fafc, duplicate I, and duplicate II in the source diagram) support standardized evaluation. Auxiliary files provide eye and mouth positions. The naming codes encode several pose orientations and acquisition categories.

4.4.2 AR Face Database

The AR-Faces slide describes 126 people and more than 4,000 colour images. For each subject, the acquisition protocol includes neutral expression, smile, anger, scream, several illumination settings, sunglasses, scarf occlusion, and a second session on a different day.

4.4.3 CMU-PIE

The CMU Pose, Illumination, and Expression database contains 68 subjects and about 608 colour images per subject, for approximately 41,368 images. The source reports 13 poses, 43 illumination conditions, and 4 expressions.

4.4.4 CASIA 3D Face V1

The lecture reports 4,624 scans of 123 people acquired with a non-contact Minolta Vivid 910 3D digitizer, including changes of viewpoint, illumination, and facial expression.

4.4.5 Labeled Faces in the Wild (LFW)

LFW contains more than 13,000 face images collected from the web. Each face is labeled with the person's name, and 1,680 individuals have two or more distinct photographs. The lecture notes that the faces were obtained in unconstrained conditions and that the dataset construction used a Viola–Jones face detector.

4.4.6 YouTube Faces

The YouTube Faces dataset contains 3,425 videos of 1,595 people, with an average of 2.15 videos per subject. The slide reports clips ranging from 48 to 6,070 frames, with an average length of 181.3 frames. Its construction started from names appearing in LFW and searched YouTube for videos of those individuals.

4.5 Structure of a Face Recognizer

The face-recognition pipeline introduced in the lecture is

1. face capture and possible image enhancement;
2. face localization, with possible extraction of one or more regions of interest (ROI) containing the whole face or components such as eyes, nose, and mouth;
3. normalization;
4. feature extraction and construction of the face template (the biometric representation used for matching).

4.6 The Face-localization Problem

Given a single image or a video sequence, face localization must determine whether one or more faces are present and estimate their positions. A useful localizer should cope with position, pose, scale, expression, illumination, and clutter; within the same image these factors can differ from subject to subject.

The lecture also uses adversarial/obfuscation examples to show that localization is not guaranteed. CV Dazzle-style makeup attempts to disrupt the structures used by face detectors, and adversarial clothing can be designed to confuse person-recognition models. The lesson treats these examples as evidence that localization is a recognition problem with failure modes, not as a trivial preprocessing step.

4.7 Families of Face-localization Methods

4.7.1 Feature-based Techniques

Feature-based approaches explicitly exploit expected facial appearance at increasing levels of structure:

- pixel-level properties such as edges and skin colour;
- facial geometry, including feature constellations and targeted feature searches;
- template matching against standard models, including correlation, snakes, and Active Shape Models in the examples shown by the lecture.

4.7.2 Image-based Techniques

Image-based techniques treat localization as a generic pattern-recognition problem with the class "face". The model learns from examples rather than requiring every facial rule to be encoded manually. The lecture illustrates subspace methods, neural networks, Support Vector Machines, and Hidden Markov Models as representative families.

4.7.3 Part-based and More Recent Approaches

The final part of Lesson 4 points to methods that localize facial parts by combining evidence from many exemplars. This local-to-global perspective is a bridge to the deep and part-response methods discussed in Lesson 5.

4.8 Chapter Summary

Face biometrics combines extremely convenient acquisition with difficult appearance variability. The recognition pipeline therefore depends critically on localization and normalization before feature extraction. Controlled face databases make nuisance factors explicit; unconstrained datasets test whether the methods generalize when those factors interact naturally.

5 More About Face Localization

Lesson 5 develops two classical localization pipelines in detail and then connects them to modern detectors. Algorithm A (Hsu–Mottaleb–Jain) is based on traditional colour/image processing. Algorithm B (Viola–Jones) uses machine learning, Haar-like features, AdaBoost, integral images, and a cascade.

5.1 Algorithm A: Hsu–Mottaleb–Jain Colour Face Detection

The algorithm is organized into two macro-phases:

1. generation of face candidates;
2. verification of the candidates through detection of facial features.

The detailed pipeline consists of illumination compensation, colour-space transformation, skin-model localization, segmentation/connected components, and localization of eyes, mouth, and the face contour.

5.1.1 Illumination Compensation

Skin tone depends on the overall scene illumination. The method uses a "reference white" formed by pixels in the top 5% of luma values, provided that a sufficient number of such pixels exists relative to image size. The lecture recalls that luma is a weighted sum of gamma-compressed RGB components, with a relation of the form

$$V_{\text{out}} = V_{\text{in}}^{\gamma}, \qquad \gamma < 1.$$

When the reference-white condition is satisfied (or the average colour is not skin-like), the RGB components are adjusted so that the average grey level of the reference white is scaled to 255.

5.1.2 Colour-space Transformation

RGB is not perceptually uniform, whereas skin detection seeks a compact cluster of similar colours. The algorithm therefore moves to a YCbCr-like representation: Y carries luminance, while C_b and C_r encode blue- and red-difference chroma. Separating luminance from chroma supports skin modelling that is less directly entangled with brightness.

5.1.3 Skin Localization and Variance-based Segmentation

After skin-colour localization, the method segments the candidate pixels and forms connected components. The lecture reviews thresholding methods including maximum entropy, Otsu's maximum-variance method, and k-means clustering.

For a grey-scale image with L grey levels, let n_i be the number of pixels with grey level i in an $M \times N$ image. Define

$$p_i = \frac{n_i}{MN}, \qquad p_i \geq 0, \qquad \sum_{i=0}^{L-1} p_i = 1.$$

For threshold t, the class probabilities are

$$\omega_0(t) = \sum_{i=0}^{t} p_i, \qquad \omega_1(t) = \sum_{i=t+1}^{L-1} p_i.$$

The class means are

$$\mu_0(t) = \frac{\sum_{i=0}^{t} i\,p_i}{\omega_0(t)}, \qquad \mu_1(t) = \frac{\sum_{i=t+1}^{L-1} i\,p_i}{\omega_1(t)},$$

and the corresponding class variances are

$$\sigma_0^2(t) = \frac{\sum_{i=0}^{t}(i-\mu_0)^2 p_i}{\omega_0}, \qquad \sigma_1^2(t) = \frac{\sum_{i=t+1}^{L-1}(i-\mu_1)^2 p_i}{\omega_1}.$$

The within-class variance used in the source is

$$\sigma_w^2(t) = \omega_0\sigma_0^2 + \omega_1\sigma_1^2,$$

and the selected threshold minimizes it:

$$t^* = \arg\min_{t\in G} \sigma_w^2(t).$$

The detected skin-tone pixels are iteratively segmented using local colour variance, converted into connected components, and grouped according to spatial closeness and similar colour to obtain face candidates.

5.2 Facial-feature Verification in Algorithm A

5.2.1 Eye Localization: Chroma and Luma Maps

The algorithm constructs two eye maps. The chrominance map uses the observation that eye regions tend to have high C_b and low C_r values:

$$\text{EyeMapC} = \frac{1}{3}\left[C_b^2 + \widetilde{C}_r^{\,2} + \frac{C_b}{C_r}\right], \qquad \widetilde{C}_r = 255 - C_r,$$

where the involved terms are normalized to $[0, 255]$ as stated in the source.

The luminance map exploits the simultaneous presence of light and dark zones around the eyes using morphological dilation and erosion with a structuring element g_σ:

$$\text{EyeMapL}(x,y) = \frac{Y(x,y) \oplus g_\sigma(x,y)}{Y(x,y) \ominus g_\sigma(x,y) + 1}.$$

The chroma map is enhanced by histogram equalization. The two maps are combined with an AND operation, after which dilation, masking, normalization, and further refinement suppress non-eye regions and emphasize the eyes.

5.2.2 Mathematical Morphology: Dilation and Erosion

Let X be the set of foreground coordinates in a binary image and K a structuring element. If K_x denotes K translated so that its centre/origin is at x, the lecture defines dilation by

$$X \oplus K = \{x \mid K_x \cap X \neq \varnothing\},$$

and erosion by

$$X \ominus K = \{x \mid K_x \subseteq X\}.$$

The structuring element determines how the foreground region grows under dilation and shrinks under erosion.

5.2.3 Mouth Localization

In the mouth region, C_r tends to be high relative to C_b. The lecture defines

$$\text{MouthMap} = C_r^2 \left(C_r^2 - \eta \frac{C_r}{C_b} \right)^2,$$

with

$$\eta = 0.95 \frac{\frac{1}{n} \sum_{(x,y) \in FG} C_r(x,y)^2}{\frac{1}{n} \sum_{(x,y) \in FG} C_r(x,y) C_b(x,y)},$$

where n is the number of pixels in the face mask and the C_r/C_b and C_r^2 responses are normalized to $[0, 255]$.

5.2.4 Face-contour Verification

Candidate triangles formed by two candidate eyes and one candidate mouth are verified using:

- luma variation and average orientation gradient in the eye/mouth blobs;
- triangle geometry and orientation;
- evidence of a face contour around the triangle.

A score is assigned to triangles satisfying the constraints, with preference for upright orientation and symmetry. The highest-scoring triangle above the acceptance threshold is selected.

5.3 Algorithm B: Viola–Jones Face Detection

Viola and Jones introduced a highly influential image-based object detector. In face localization it learns a binary classifier for *face* versus *non-face*. Training is relatively expensive, but the detection procedure is designed to be extremely fast.

The key ideas emphasized by the lecture are:

- integral images for fast feature evaluation;
- boosting for selecting discriminative features and combining weak learners;
- multiscale detection with a sliding window;
- a cascade that rejects easy negative windows as early as possible.

5.3.1 AdaBoost and Weak Learners

Boosting combines M weak classifiers $h_i(x)$ into a stronger nonlinear ensemble. The lecture writes the normalized weighted combination as

$$H_M(x) = \frac{\sum_{i=1}^{M} \alpha_i h_i(x)}{\sum_{i=1}^{M} \alpha_i},$$

where $h_i(x) \in \{-1, +1\}$ and $\alpha_i \geq 0$ is the relative weight of the i-th weak learner.

AdaBoost learns both the weak classifiers and their weights from training pairs $\{(x_i, y_i)\}_{i=1}^{N}$, with $y_i \in \{-1, +1\}$. Training maintains weights on the examples. Samples that are difficult to classify receive greater weight in later rounds, concentrating subsequent weak learners on earlier errors.

5.3.2 Haar-like Rectangle Features

For images, Viola–Jones uses simple rectangular features. The feature value is the difference between the sum of pixel intensities in designated white and black rectangular regions. A 24×24 window already permits on the order of 180,000 possible rectangular features, so evaluating every feature at test time would be impractical. AdaBoost selects a small, discriminative subset.

5.3.3 Integral Images

The integral image at (x, y) is

$$II(x, y) = \sum_{x' \le x,\, y' \le y} I(x', y').$$

It permits the sum of pixels inside any axis-aligned rectangle to be obtained from four integral-image values. Using the corner labels from the lecture,

$$\text{sum}(D) = II(4) - II(2) - II(3) + II(1).$$

This constant-time rectangle evaluation is what makes large numbers of Haar features computationally feasible during scanning.

5.3.4 Weak and Strong Classifiers

A one-node weak classifier compares a selected feature z_k to a threshold t_k. For the M-th weak learner the lecture uses the form

$$h_M(x) = \begin{cases} +1, & z_{k^*} > t_{k^*}, \\ -1, & \text{otherwise.} \end{cases}$$

For each candidate feature, a threshold is chosen to minimize classification error; the feature/threshold pair with the lowest error becomes the next weak classifier. AdaBoost then combines the selected weak classifiers into the strong classifier.

5.3.5 Cascade of Classifiers

The cascade starts with simple stages that reject many negative subwindows while retaining almost all positives. Only windows that pass one stage are sent to the next, more complex stage. A negative result at any stage immediately rejects the window.

The training example in the source illustrates cumulative false-positive reduction: a one-feature stage may keep a 100% detection rate with roughly 50% false positives; a five-feature stage with 40% false positives gives about 20% cumulative false positives; a later twenty-feature stage at 10% reduces the cumulative value to about 2%.

After a face ROI has been detected, the same general machinery can be applied hierarchically with models trained for eyes, nose, and mouth.

5.4 Deep and Modern Face-localization Approaches

The lecture then moves from hand-designed cascades to learned representations. One illustrated deep approach scores *face likeliness* from local facial-part responses: part-level response maps are generated across the full image and are then aggregated from local evidence to a global face hypothesis.

5.4.1 dlib

dlib is presented with two built-in face detectors:

- a HOG (Histogram of Oriented Gradients) plus linear SVM detector, computationally efficient but less robust to large pose changes, occlusion, or very small faces;
- a max-margin (MMOD) CNN detector, described in the lecture as more robust to viewing angle, illumination, and occlusion.

The library also provides a pre-trained 68-point facial-landmark detector.

5.4.2 MediaPipe

The MediaPipe face detector operates on images or video streams and outputs face locations together with a small set of facial key points. The slide lists the left eye, right eye, nose tip, mouth, left eye tragion, and right eye tragion.

5.4.3 YOLO

YOLO (You Only Look Once) is introduced as a neural-network architecture for object detection that can also be adapted to faces. The key point in the lecture is the shift from a hand-crafted face-specific pipeline to a general learned object-detection architecture.

5.5 Evaluating Face Localization

The lecture lists three practical error quantities:

- **false positives**: percentage of windows classified as faces although they contain no face;
- **not localized faces**: percentage of true faces that the system fails to identify;
- **C-Error**: localization error defined from the Euclidean distance between the true face centre and the estimated centre, normalized with respect to the sum of the axes of the ellipse containing the face.

Thus detector evaluation must account for both *whether* a face is found and *where* it is localized.

5.6 Chapter Summary

The two classical algorithms illustrate complementary design philosophies. Hsu–Mottaleb–Jain uses domain knowledge about colour, morphology, and facial geometry. Viola–Jones learns a discriminative cascade over simple Haar-like features and makes evaluation fast with integral images. Modern methods retain the same localization objective but learn increasingly rich representations and part/object relationships with HOG/SVM, CNN, MediaPipe-style models, or YOLO-like architectures.

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Quick Review

The most exam-relevant relationships are grouped below for rapid review.

Key Concept

Foundations and architecture

- Biometrics is pattern recognition in which the classes are individual people.
- A biometric system uses acquisition, feature extraction, matching, and decision modules.
- Enrollment creates reference templates; recognition compares a probe with stored templates.
- Verification is 1:1; identification is 1:N.
- A useful trait should provide universality, uniqueness, permanence, collectability, and acceptability.
- Controlled acquisition reduces nuisance variation; uncontrolled acquisition may not permit recapture.

Remark – Matching rules

$$\text{similarity: accept} \iff s \geq t,$$
$$\text{distance: accept} \iff d \leq t.$$

Increasing a *similarity* threshold makes acceptance stricter; increasing a *distance* threshold makes acceptance looser.

Performance Metric – Verification

- FAR/FMR: impostor claim incorrectly accepted.
- FRR/FNMR: genuine claim incorrectly rejected.
- GAR/GMR: $1 - \text{FRR}$.
- EER: operating point where FAR and FRR are equal.
- Genuine and impostor score distributions overlap; the threshold divides that overlap into false acceptances and false rejections.
- ROC plots GAR versus FAR. DET plots FRR versus FAR on normal-deviate (probit) axes.

Performance Metric – Identification

- Open set asks both whether the probe belongs to the gallery and, if so, who it is.
- $\text{DIR}(t, k)$: successful detection with the correct identity within the first k ranks.
- $\text{FNIR}(t) = 1 - \text{DIR}(t, 1)$.
- FPIR: false alarm probability for non-gallery probes.
- Closed set assumes every probe has a gallery mate and is evaluated by rank.
- CMS@k: probability that the correct identity is within the first k candidates.
- CMC plots CMS versus rank; CMS@1 is the Recognition Rate.

Remark – Do not confuse

- FAR and FRR are not precision and recall. FAR corresponds to FPR/fall-out; FRR corresponds to FNR/miss rate.
- Type I/II labels depend on the null hypothesis. FAR/FRR terminology is safer.
- Standard DET axes are probit, not ordinary logarithmic axes.
- EER is a comparison point, not a universally optimal deployment threshold.
- A raw sample is distinct from the feature representation or template extracted from it.

Key Concept

Modalities and representations

- Strong biometric traits aim at identity discrimination; soft traits mainly narrow the candidate set.
- Vector templates may use Euclidean distance or cosine similarity; histograms/distributions may use correlation or Bhattacharyya distance; time series may use DTW.
- Learned embeddings can be compared as vectors; structured templates such as fingerprint minutiae require task-specific correspondence.
- Re-identification is ranked retrieval across observations: CMC is useful for a single mated gallery image, while AP/mAP address multiple relevant matches.

Remark – Compact equation sheet

Verification	$\mathrm{FAR} = \dfrac{FA}{FA+GR}, \quad \mathrm{FRR} = \dfrac{FR}{GA+FR}, \quad \mathrm{GAR} = 1 - \mathrm{FRR}.$
Equal error	$\mathrm{FAR}(t^*) = \mathrm{FRR}(t^*)$.
Open set	$\mathrm{FNIR}(t) = 1 - \mathrm{DIR}(t, 1)$; FPIR is the fraction of non-gallery probes whose best gallery score reaches the threshold.
Closed set	$\mathrm{RR} = \mathrm{CMS@1}$; CMS@$k$ is the probability that the correct identity is within the first k candidates.

Key Concept

Recognition reliability (Lesson 3)

- Global FoMs do not tell us whether a particular response is trustworthy.
- Dataset composition, probe/gallery size, acquisition variation, and cross-dataset behaviour matter when comparing systems.
- Doddington's menagerie describes subject-specific genuine/impostor score behaviour (sheep, goats, lambs, wolves), extended by chameleons, phantoms, doves, and worms.
- Quality evaluates the input; response reliability evaluates a system decision/output.
- SRR measures candidate confusion for one probe using relative-distance or density-ratio evidence and a critical reliability value.
- Template updating addresses ageing/enhancement by incorporating operational data into gallery representations.

Key Concept

Face recognition and localization (Lessons 4–5)

- Face is highly acceptable and easy to capture, but strongly affected by PIE (pose, illumination, expression) and ageing.
- Recognition pipeline: capture/enhancement → localization/ROI → normalization → feature extraction/template.
- Feature-based localization exploits edges, skin colour, geometry, and templates; image-based localization learns the face class from examples.
- Hsu–Mottaleb–Jain: illumination compensation, YCbCr/skin model, segmentation, connected components, eye/mouth maps, morphology, facial geometry.
- Viola–Jones: Haar-like features + integral images + AdaBoost + multiscale sliding windows + cascade.
- Modern detectors include HOG/SVM and CNN approaches, MediaPipe-style landmark outputs, and YOLO-like object detectors.
- Localization evaluation includes false positives, missed faces, and centre-localization error (C-Error).

Remark – Additional equations

$$M(\Delta) = |\mathrm{FAR}(\Delta) - \mathrm{FRR}(\Delta)|, \qquad \varphi_{\mathrm{DR}}(p) = 1 - \frac{|N_b|}{|G|}.$$

$$II(x,y) = \sum_{x' \le x,\, y' \le y} I(x',y'), \qquad X \oplus K = \{x : K_x \cap X \neq \varnothing\}, \qquad X \ominus K = \{x : K_x \subseteq X\}.$$